

Review IV

1. Section 5.2
 - a. Approximating area under a curve $y = f(x)$ over the interval $I = [a, b]$
 - i. Partition the interval I uniformly into n subintervals of equal width
 - ii. Compute the common uniform width $\Delta x = (b-a)/n$ of the subintervals
 - iii. Select the right-hand node point x_k for each subinterval
 - iv. For each subinterval compute the rectangular area $f(x_k)\Delta x$
 - v. Sum up the rectangular areas
 - b. Summation notation and properties
 - i. Basic rules: Theorem 5.4
 - ii. Summation Formulas: Page 334
 - c. Representative problems: 4, 8, 9, 11, 15, 24
2. Section 5.3
 - a. Riemann sum for $y = f(x)$ over the interval $I = [a, b]$
 - i. Partition the interval I into n subintervals $I_k, 1 \leq k \leq n$
 - ii. Compute the width of each subinterval, $\Delta x_k = x_k - x_{k-1}$
 - iii. Select a node point x_k^* in each subinterval I_k
 - iv. For each subinterval I_k compute the product $f(x_k^*)\Delta x_k$
 - v. Sum up the products
 - b. Definition of definite integral: limit process of Riemann sums
 - c. Notation $\int_a^b f(x)dx$
 - d. Theorem 5.5 If f is continuous on $[a, b]$, then $\int_a^b f(x)dx$ exists.
 - e. Area under a curve $y = f(x)$ over the interval $I = [a, b]$ as an integral $\int_a^b f(x)dx$.
 - f. Properties of definite integrals: Theorem 5.6
 - g. Representative problems: 3, 9, 10, 18, 19, 26a
3. Section 5.4
 - a. First Fundamental Theorem of Calculus: Theorem 5.7
 - b. Second Fundamental Theorem of Calculus: Theorem 5.8
 - c. Representative problems: 6, 12, 18, 26, 28, 32, 35, 38, 41, 43
4. Section 5.5
 - a. Examples: 2, 3, 4, 8
 - b. Representative problems: 3, 6, 8, 15, 18, 24, 29, 33, 35, 42, 46

5. Section 5.6
 - a. Definition of differential equation
 - b. Direction fields
 - c. First-order, separable differential equations
 - d. Exponential growth/decay equation
 - e. Orthogonal trajectories
 - f. Representative problems: 10, 12, 23, 26, 30, 37, 60, 65

6. Section 5.7
 - a. Mean Value Theorem for integrals
 - b. Average value of $y = f(x)$ over the interval $I = [a, b]$
 - c. Representative problems: 3, 8, 18, 21, 25, 35

7. Section 5.8
 - a. Integral approximation by Riemann sum
 - b. Integral approximation by Trapezoid Rule
 - c. Integral approximation by Simpson's Rule
 - d. Error estimates for Trapezoid Rule and Simpson's Rule
 - e. Representative problems: 3, 7, 12, 13, 19